CS565 - Intro to Artificial Intelligence Project 1: Battleship

Due: February 24, 2025

Introduction

Files to Edit and Submit

You will fill in portions of the provided Python files during the assignment. You should submit these files with your code and comments. **Please do not change the other files in this distribution** or submit any of the original files other than the ones explicitly required.

Evaluation

Your code will be autograded for technical correctness. **Do not change the names of any provided functions or classes** within the code, as this will interfere with the test cases. There will be test cases provided for you to evaluate your implementation. However, **not all test cases will be shown**; additional hidden test cases will be used during grading to assess the robustness and correctness of your code. You are encouraged to write your own test cases to thoroughly test your implementation and ensure it handles edge cases and unexpected inputs correctly. The correctness of your implementation—not the test case's results—will ultimately determine your score. If necessary, we will manually review and grade submissions to ensure fairness.

Academic Dishonesty

We will be checking your code against other submissions in the class for logical redundancy. If you copy someone else's code and submit it with minor changes, **we will know**. Our detection tools are highly effective, so please avoid attempting this. We trust you to submit your own work—please do not let us down. Any confirmed instances of academic dishonesty will result in the **strongest consequences available**.

Getting Help

You are not alone! If you find yourself stuck, please reach out to the course staff. **Office hours, section, and the discussion forum** are all available for your support. If you can't make it to office hours, let us know, and we will arrange additional times.

We aim for these projects to be **rewarding and educational**, not frustrating or demoralizing. If you need help, ask—we are here to support you.

Discussion

Please avoid posting spoilers or sharing solutions in public forums. Discussions should focus on **concepts and strategies** rather than explicit answers. Let's maintain an environment where everyone can learn and succeed!

Missions

In the year 2045, naval warfare and maritime navigation have entered a new era. A coalition of nations has developed a cutting-edge **LIDAR Detection System** (LDS) to manage coastal defense, track naval fleets, and navigate treacherous waters. The LDS is tasked with analyzing vast grids of maritime zones, identifying battleship formations, detecting clear navigation paths, and solving logistical challenges of fleet positioning. Below is a breakdown of how the system operates in four critical mission scenarios:

Mission 1: Counting Battleships (Detection of Enemy Fleets)

The LIDAR system scans a maritime grid, where: - **Battleships** are represented as '1'. - **Open water** is represented as '0'.
Using **Depth First Search (DFS)**, the system identifies clusters of battleships connected horizontally or vertically. This information is critical for understanding enemy fleet organization.

Mission 2: Navigating the Grid (Clearing a Path to Safety)

The coalition's ships often find themselves needing to traverse hostile waters filled with enemy battleships. These waters are represented as a square grid where: - '0' denotes **open water**. - '1' denotes **battleships blocking the path**.
The LIDAR system deploys **Breadth First Search (BFS)** to calculate the shortest path through open water, avoiding battleships. Paths are connected 8-directionally, and the LIDAR alerts commanders if no path exists.

Mission 3: Crossing Treacherous Waters (Minimizing Effort in Difficult Terrain)

Not all challenges are caused by enemy battleships; nature itself poses significant obstacles. The LIDAR system scans a grid of **elevation data**, where higher values indicate more difficult terrain, such as strong currents, rocky shoals, or shallow waters.

Using **Uniform Cost Search (UCS)**, the system determines the minimum effort path, defined as the route with the smallest maximum elevation difference between any two cells.

Mission 4: Fleet Maneuvering (Arranging Battleships into Strategic Formations)

The LIDAR organizes a 2x3 grid of battleships and open water to match the target formation:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 0 \end{bmatrix}$$

Using **A* Search**, the system calculates the optimal sequence of moves to arrange the fleet. Battleships can only move into adjacent open water cells. If the formation is impossible, the system provides an alert.